JAX LaTeX: Isn't It Nice To Have Something Like This?

George Snell*, Douglass Coleman*†, Muerial Davisson, Jane Austin, Roscoe Jackson‡

The Jackson Laboratory
600 Main Street, Bar Harbor, ME 04609, USA

Abstract

Your abstract goes here. Summarize the purpose, methods, key findings, and conclusions of your work.

***Equal contribution, †Correspondence, ‡Senior author**
Code: https://github.com/kumarlabjax
Correspondence: vivek.kumar@jax.org
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1 Introduction

Protein folding is one of the most challenging problems in computational biology. The ability to predict protein structures from amino acid sequences has profound implications for drug discovery, disease understanding, and biotechnology applications.

Recent advances in deep learning have revolutionized protein structure prediction, with methods like AlphaFold [2] achieving remarkable accuracy. Similarly, RoseTTAFold [1] has demonstrated the power of three-track neural networks in this domain.

2 Methods

2.1 Data Collection

Describe your data collection methods here. Include details about:

- Data sources
- Sample size
- Inclusion/exclusion criteria

- Data preprocessing steps

2.2 Experimental Design

Describe your experimental design here. Include:

- Study design type
- Control groups
- Randomization procedures
- Blinding methods

2.3 Statistical Analysis

Describe your statistical methods here. Include:

- Statistical tests used
- Software packages
- Significance levels
- Multiple comparison corrections

3 Results

3.1 Primary Findings

3.2 Secondary Analysis

Present additional analyses here. This could include:

- Sensitivity analyses
- Subgroup analyses
- Exploratory findings
- Validation results

4 Discussion

4.1 Interpretation of Results

- Previous research

- Theoretical frameworks
- Clinical or practical implications
- Biological significance

4.2 Limitations

Discuss any limitations of your study. Be honest about:

- Sample size limitations
- Methodological constraints
- Potential biases
- Generalizability issues

4.3 Future Directions

Discuss future research directions. This could include:

- Follow-up studies
- Methodological improvements
- Clinical applications
- Theoretical developments

5 Supplement

5.1 Additional Figures

Include supplementary figures here. These are figures that support your main findings but are not essential for the main narrative.

5.2 Additional Tables

Include supplementary tables here. These provide additional data that supports your main findings.

5.3 Additional Methods

Include additional methodological details here. This is where you can provide:

- Detailed protocols

Parameter	Group A	Group B	Group C
Mean	10.5	12.3	9.8
SD	2.1	1.9	2.3
Median	10.2	12.0	9.5
Range	5.1–15.9	8.4–16.2	5.3–14.3

Table 1: Supplementary Table 1: Descriptive statistics by group.

- Additional validation procedures
- Computational details
- Statistical code

6 References

References

[1] Minkyung Baek, Frank DiMaio, Ivan Anishchenko, Justas Dauparas, Sergey Ovchinnikov, Gyu Rie Lee, Jue Wang, Qian Cong, Lisa N Kinch, R Dustin Schaeffer, et al. Accurate prediction of protein structures and interactions using a three-track neural network. *Science*, 373(6557):871–876, 2021.

[2] John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873):583–589, 2021.